# Q1 AI Tools Declaration

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I used the following AI tools while preparing this submission.

| Tool | Purpose | What the tool did | My own contribution / verification |
|---|---|---|---|
| Claude Code (Anthropic), Claude Opus models | Research and data collection | Searched provider documentation, pricing and rate-limit pages, the OpenRouter and Hugging Face APIs and Artificial Analysis, and compiled data/models.csv and data/sources.csv with a source tag for every value. | I took the Artificial Analysis leaderboard screenshots that the AutomationBench-AA and Terminal-Bench 4.0 values come from, and the Hugging Face config.json screenshot in Figure 2. |
| Claude Code | Code | Wrote latency_logger.py (OpenAQ, Open-Meteo, heat-index and knowledge-base tools plus the timing logger), the notebook generator build/build_q1.py and the report builder build/build_report.py. | I ran the latency logger with my own OpenAQ key and checked that the timing log covers two days. |
| Claude Code | Calculations and analysis | Implemented the Section 2 formulas (token volume, caching cost, peak load, KV cache, break-even, latency) in the notebook following the assignment's assumptions. | I checked the parameter derivation from my roll number and the Section 2 results in the report. |
| Claude Code | Writing | Drafted the report text and tables from the notebook outputs, rewrote them in plain prose, applied the document formatting, and drafted this declaration. | I had the text rewritten in plain prose, set the document formatting and reviewed the final PDF. I agree with the conclusions and recommendation and can defend them. |

All numbers in the report are produced by the submitted notebook from the submitted data files and timing log, and every external figure is cited to its source. I understand the work and take responsibility for its correctness.

Signature: ukkashah Date: 10 November 2026